

PathFinder — Database Design Document

1. Database Overview

PathFinder uses **SQLite** as its relational database, managed through **Prisma ORM**. SQLite was chosen for its zero-configuration setup, portability (single file), and suitability for development and light production workloads.

The schema is defined in `backend/prisma/schema.prisma` and consists of **16 interconnected tables** that model the complete domain of the application.

2. Entity-Relationship Diagram (Textual)

User (1)	— (0..1)	StudentProfile
User (1)	— (0..1)	OrganizationProfile
User (1)	— (0..N)	Notification
User (1)	— (0..N)	Bookmark
User (1)	— (0..N)	AIGeneration
User (1)	— (0..N)	SavedSearch
User (1)	— (0..N)	AuditLog
User (1)	— (0..N)	Ticket
User (1)	— (0..N)	TicketMessage
StudentProfile (1)	— (0..N)	Application
StudentProfile (1)	— (0..N)	Document
StudentProfile (1)	— (0..N)	Skill
StudentProfile (1)	— (0..N)	Certification
StudentProfile (1)	— (0..N)	PortfolioLink
OrganizationProfile (1)	— (0..N)	Opportunity
Opportunity (1)	— (0..N)	Application
Opportunity (1)	— (0..N)	Bookmark
Application (1)	— (0..N)	ApplicationEvent
Ticket (1)	— (0..N)	TicketMessage

3. Table Definitions

3.1 User

The central authentication table. Every person on the platform has exactly one User record.

Column	Type	Constraints	Description
id	String (UUID) PK, auto-generated	Unique	Unique identifier
email	String	UNIQUE, NOT NULL	Login email address
password	String	NOT NULL	bcrypt-hashed password
role	String	NOT NULL	'STUDENT', 'ORGANIZATION', or 'ADMIN'
createdAt	DateTime	DEFAULT now()	Registration timestamp
updatedAt	DateTime	Auto-updated	Last modification timestamp
resetToken	String?	NULLABLE	Secure token for password reset
resetTokenExpiry	DateTime?	NULLABLE	Expiry time for reset token (1 hour)

Relationships: Has 0-or-1 StudentProfile, 0-or-1 OrganizationProfile, many Notifications, Bookmarks, AIGenerations, SavedSearches, AuditLogs, Tickets, and TicketMessages.

3.2 StudentProfile

Extended profile information specific to student users. One-to-one with User.

Column	Type	Constraints	Description
id	String (UUID) PK	Unique	Unique identifier
userId	String	UNIQUE, FK → User.id	Link to parent User

firstName	String	NOT NULL	Student's first name
lastName	String	NOT NULL	Student's last name
bio	String?	NULLABLE	Short biography
major	String?	NULLABLE	Academic major
university	String?	NULLABLE	University name
graduationYear	Int?	NULLABLE	Expected graduation year
createdAt	DateTime	DEFAULT now()	Creation timestamp
updatedAt	DateTime	Auto-updated	Last update timestamp

Cascade Delete: When a User is deleted, their StudentProfile is automatically deleted.

3.3 OrganizationProfile

Extended profile for organization users. One-to-one with User.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
userId	String	UNIQUE, FK → User.id	Link to parent User
companyName	String	NOT NULL	Organization name
description	String?	NULLABLE	Company description
logoUrl	String?	NULLABLE	URL to uploaded logo
website	String?	NULLABLE	Company website URL
industry	String?	NULLABLE	Industry sector
createdAt	DateTime	DEFAULT now()	Creation timestamp
updatedAt	DateTime	Auto-updated	Last update timestamp

3.4 Opportunity

Represents a single internship, scholarship, grant, or hackathon listing.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
orgId	String	FK → OrganizationProfile.id	Creating organization
title	String	NOT NULL	Opportunity title
description	String	NOT NULL	Full description
category	String	NOT NULL	'INTERNSHIP', 'SCHOLARSHIP', 'GRANT', 'HACKATHON'
deadline	DateTime?	NULLABLE	Application deadline
eligibility	String?	NULLABLE	Eligibility criteria
status	String	DEFAULT 'PUBLISHED'	'DRAFT', 'PUBLISHED', 'ARCHIVED'
isFeatured	Boolean	DEFAULT false	Featured listing flag
location	String?	NULLABLE	'Remote', 'On-site', city name
stipend	String?	NULLABLE	Compensation details
duration	String?	NULLABLE	Duration of opportunity
createdAt	DateTime	DEFAULT now()	Creation timestamp
updatedAt	DateTime	Auto-updated	Last update timestamp

3.5 Application

The join table between StudentProfile and Opportunity. Represents a single application.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
opportunityId	String	FK → Opportunity.id	Target opportunity
studentId	String	FK → StudentProfile.id	Applying student
status	String	DEFAULT 'PENDING'	Application status
coverLetter	String?	NULLABLE	Student's cover letter
answers	String?	NULLABLE	JSON string for custom form answers
notes	String?	NULLABLE	Internal notes from organization
withdrawnAt	DateTime?	NULLABLE	Withdrawal timestamp
matchScore	Int?	NULLABLE	AI-calculated match score (0-100)
matchReason	String?	NULLABLE	Human-readable match explanation
createdAt	DateTime	DEFAULT now()	Submission timestamp
updatedAt	DateTime	Auto-updated	Last update timestamp

Application Statuses: PENDING → REVIEWING → SHORTLISTED → ACCEPTED / REJECTED / WITHDRAWN

3.6 ApplicationEvent

Timeline entries tracking the lifecycle of each application.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
applicationId	String	FK → Application.id	Parent application
type	String	NOT NULL	'SUBMITTED', 'STATUS_CHANGED', 'WITHDRAWN', 'NOTE_ADDED'
details	String?	NULLABLE	Description of the event
createdAt	DateTime	DEFAULT now()	Event timestamp

3.7 Skill

Individual skills associated with a student profile. Many-to-one with StudentProfile.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
name	String	NOT NULL	Skill name (e.g., 'React', 'Python')
studentProfileId	String	FK → StudentProfile.id	Owning student

3.8 Certification

Professional certifications held by a student.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
name	String	NOT NULL	Certificate name
issuer	String	NOT NULL	Issuing organization
dateIssued	String?	NULLABLE	Date of issuance
studentProfileId	String	FK → StudentProfile.id	Owning student

3.9 PortfolioLink

External portfolio links (GitHub, LinkedIn, personal website).

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
title	String	NOT NULL	Link label (e.g., 'GitHub')
url	String	NOT NULL	Full URL
studentProfileId	String	FK → StudentProfile.id	Owning student

3.10 Document

Uploaded files associated with a student (resumes, SOPs, transcripts).

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
studentId	String	FK → StudentProfile.id	Owning student
title	String	NOT NULL	Document title
fileUrl	String	NOT NULL	URL path to the uploaded file
type	String	NOT NULL	'RESUME', 'SOP', 'TRANSCRIPT', 'OTHER'
createdAt	DateTime	DEFAULT now()	Upload timestamp
updatedAt	DateTime	Auto-updated	Last update timestamp

3.11 Bookmark

Saved opportunities. Composite unique constraint prevents duplicate bookmarks.

Column	Type	Constraints	Description
id	String (UUID) PK		Unique identifier
userId	String	FK → User.id	Bookmarking user
opportunityId	String	FK → Opportunity.id	Bookmarked opportunity
createdAt	DateTime	DEFAULT now()	Bookmark timestamp

Unique Constraint: @@unique([userId, opportunityId])

3.12 Notification

In-app notifications for user events.

Column	Type	Constraints	Description
id	String (UUID)	PK	Unique identifier
userId	String	FK → User.id	Recipient user
title	String	DEFAULT ""	Notification title
message	String	NOT NULL	Notification body
type	String	DEFAULT "SYSTEM"	'SYSTEM', 'STATUS_CHANGE', 'NEW_APPLICATION', 'DEADLINE', 'NEW_OPPORTUNITY'
link	String?	NULLABLE	Optional navigation link
read	Boolean	DEFAULT false	Read/unread status
createdAt	DateTime	DEFAULT now()	Creation timestamp

3.13 AIGeneration

History of all AI-generated content.

Column	Type	Constraints	Description
id	String (UUID)	PK	Unique identifier
userId	String	FK → User.id	Generating user
type	String	NOT NULL	'SOP', 'COVER_LETTER', 'ESSAY', 'PERSONAL_STATEMENT'
prompt	String	NOT NULL	Input prompt or context
output	String	NOT NULL	Generated content
oppld	String?	NULLABLE	Related opportunity (if any)
createdAt	DateTime	DEFAULT now()	Generation timestamp

3.14 SavedSearch

Saved search parameters for quick re-execution.

Column	Type	Constraints	Description
id	String (UUID)	PK	Unique identifier
userId	String	FK → User.id	Owning user
name	String	NOT NULL	Search name
query	String	NOT NULL	JSON stringified search parameters
createdAt	DateTime	DEFAULT now()	Save timestamp

3.15 AuditLog

Platform-wide audit trail for compliance and debugging.

Column	Type	Constraints	Description
id	String (UUID)	PK	Unique identifier
userId	String?	FK → User.id (optional)	Acting user
action	String	NOT NULL	Action type (e.g., 'USER_LOGIN')
entity	String?	NULLABLE	Entity type affected
entityId	String?	NULLABLE	Entity ID affected
details	String?	NULLABLE	Additional details
ip	String?	NULLABLE	Client IP address
createdAt	DateTime	DEFAULT now()	Action timestamp

3.16 Ticket & TicketMessage

Support ticketing system with threaded conversations.

Ticket Table:

Column	Type	Constraints	Description
id	String (UUID)	PK	Unique identifier
subject	String	NOT NULL	Ticket subject line
description	String	NOT NULL	Detailed description
status	String	DEFAULT 'OPEN'	'OPEN', 'IN_PROGRESS', 'RESOLVED', 'CLOSED'

type	String	DEFAULT 'OTHER'	'BUG', 'REPORT', 'FEATURE_REQUEST', 'OTHER'
priority	String	DEFAULT 'MEDIUM'	'LOW', 'MEDIUM', 'HIGH'
creatorId	String	FK → User.id	Ticket creator
createdAt	DateTime	DEFAULT now()	Creation timestamp
updatedAt	DateTime	Auto-updated	Last update timestamp

TicketMessage Table:

Column	Type	Constraints	Description
id	String (UUID)	PK	Unique identifier
ticketId	String	FK → Ticket.id	Parent ticket
senderId	String	FK → User.id	Message author
message	String	NOT NULL	Message content
createdAt	DateTime	DEFAULT now()	Send timestamp

4. Cascade Delete Rules

All foreign key relationships use `onDelete: Cascade`, ensuring:

- Deleting a User automatically deletes their profile, applications, notifications, bookmarks, tickets, and all related records.
- Deleting an Opportunity automatically deletes all associated applications and bookmarks.
- Deleting a Ticket automatically deletes all associated messages.

5. Naming Conventions

- **Tables:** PascalCase (e.g., StudentProfile, ApplicationEvent)
- **Columns:** camelCase (e.g., firstName, matchScore, createdAt)
- **Primary Keys:** Always named `id`, type UUID string
- **Foreign Keys:** Named as `<relation>Id` (e.g., userId, opportunityId, studentId)
- **Timestamps:** `createdAt` (auto-set), `updatedAt` (auto-updated)
- **Booleans:** Descriptive names (e.g., read, isFeatured)

6. Indexing

SQLite automatically indexes:

- All primary key columns
- All columns with `@unique` constraints
- Composite unique constraints (e.g., Bookmark's [userId, opportunityId])

End of Database Design Document